Using inequality notation for errors in calculations

1 ___________ are the digits in a number that contribute to the accuracy of the number. The first significant figure is the first non-zero digit. Tick **1** correct answer

- ☐ Estimated numbers
- ☐ Important numbers
- ☐ Real numbers
- ✓ Significant figures

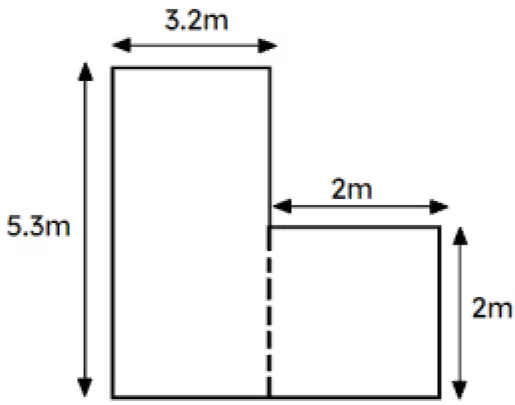
2 When a number, x, is rounded to the nearest integer the result is 35. Select the numbers which complete the error interval for x: $\square \leq x < \square$ Tick **1** correct answer

- ☐ 35 and 45
- ✓ 34.5 and 35.5
- ☐ 34.999 and 35.999
- ☐ $34.4\dot{9}$ and $35.4\dot{9}$

3 A long road is made up of 3 sub roads. The 1st road measures 35 m (2 s.f), the 2nd road is 41 m (2 s.f) and the 3rd road is 48 m (2 s.f). Find the error interval for the length x of the long road. Tick **1** correct answer

- ✓ $122.5 \leq x < 125.5$
- ☐ $122 \leq x < 125$
- ☐ $122.5 \leq x < 125.59$
- ☐ $124.5 \leq x < 126.5$

4 A wall is in an L-shape. All the lengths are given to 1 d.p or the nearest integer. Work out the error interval for the area of the wall. Tick **1** correct answer



- ☐ $19.6375 \leq x < 22.7875$
- ☐ $18 \leq x < 23$
- ☐ $18.96 \leq x < 20.96$
- ☑ $18.7875 \leq x < 23.6375$

5 A square has lengths in cm given to 1 decimal place. The error interval of the perimeter of the square is $31 \leq x < 31.4$. Work out the length of the square correct to 1 d.p. Tick **1** correct answer

- ☐ 7.2
- ☐ 7.4
- ☐ 7.6
- ☑ 7.8
- ☐ 8.0

6 A square has lengths in cm to 1 decimal place. The error interval of the area of the square is $30.8025 \leq x < 31.9225$. Work out the length of the square to 1 d.p. Tick **1** correct answer

- ☐ 5.5 cm
- ☐ 5.4 cm
- ☑ 5.6 cm
- ☐ 5.8 cm
- ☐ 5.3 cm